Reaching Every Residue: Backward Trees of $qx+1$ Maps with Non-Wieferich Multiplier

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Abstract

Let $q > 1$ be odd with $\gcd\bigl(q, (2^{\mathrm{ord}_q(2)} - 1)/q\bigr) = 1$, that is, not a Wieferich number in the sense of Franco and Pomerance, and let Col_q halve even numbers and send odd x to $qx+1$. Let $t \geq 1$ be congruent modulo q to a power of 2, and let $L = \lceil \log_2 q \rceil$. We prove that for every modulus m, residue r and threshold B some $w \equiv r \pmod m$ with $B < w \leq (B+1)\, t\, 2^{2m^{L+1}q^m}$ reaches t. So the predecessors of t meet every residue class modulo every m, effectively; equivalently, they are dense in the profinite integers. The hypothesis on q cannot be dropped: for odd q, the predecessors of 1 meet every residue class modulo every m if and only if q is not a Wieferich number, and for a Wieferich number they stay inside a proper union of residue classes that is closed under taking predecessors. For $5x+1$, whose known cycles through 1, 13 and 17 have pairwise disjoint backward trees, each of the three trees meets every residue class. When every odd prime factor of m divides q, the proof gives the sharper bound $(B+1)\, t\, 2^{\,d\, m^{\log_2 q} + O(\log m)}$ with $d = \mathrm{ord}_q(2)$, and for powers of q a single exponential $(B+1)\, t\, 2^{\,d(m+1)}$. Theorems A and B are also verified in Lean v4.35.0. The paper makes no claim about the Collatz conjecture.

Keywords: Collatz problem; $3x+1$ problem; $qx+1$ maps; generalized Collatz maps; backward tree; predecessor set; residue classes; profinite integers; Wieferich numbers; Crandall numbers; Lean 4; formal verification.

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1 Introduction

For an integer $q > 1$ define $\mathrm{Col}_q : \mathbb{N} \to \mathbb{N}$ by

$$\mathrm{Col}_q(x) = \begin{cases} x/2 & x \text{ even}, \\ qx + 1 & x \text{ odd}. \end{cases}$$

Here $\mathbb{N} = \{0, 1, 2, \dots\}$; since Col_q maps positive integers to positive integers, and all statements below concern positive integers, they may be read over the positive integers. We say x *reaches* y, written $x \to y$, if $\mathrm{Col}_q^k(x) = y$ for some $k \geq 0$. For a *target* t, the set of x reaching t is the *predecessor set* of t [Wir98]; we sometimes call it the backward tree of t, rooted at t, which Applegate and Lagarias call the tree of preimages [AL95b]. An element of it with prescribed properties is a *witness*, as in Theorem A. Throughout, q is not assumed prime; the letter p is reserved for primes.

Notation. The multiplier is called q as in Crandall, Steiner and Franco–Pomerance [Cra78, Ste81, FP95]. The map Col_q is the unaccelerated map, written C by Lagarias and Col by Tao [Lag85, Tao22]; we follow Tao. Much of the literature instead uses the accelerated map T_q (written T by Lagarias), which sends odd x to $(qx + 1)/2$ and whose orbits omit the value $qx + 1$ after each odd x, or the odd-to-odd map C_q of Crandall and Franco–Pomerance, which sends odd x to the largest odd divisor of $qx + 1$. The maps S_0, S_1 of §6 are the two branches of T_q. We work with the unaccelerated map because the statements concern it: predecessor sets are closed under doubling, the targets in Theorem A may be even, and the residue classes of even numbers are included.

Many results on predecessor sets concern counting: how many $y \leq x$ reach 1 [AL95, KL03], or what proportion of orbits behave well [Tao22]. Here we ask which residue classes a predecessor set meets. For $3x + 1$ the answer is essentially known. Monks [Mon06] proved that every arithmetic progression is *sufficient*: every orbit merges with the orbit of some element of it. His theorem completed a line of partial results on particular progressions, reported to include Cadogan [Cad84] (the class 1 modulo 4), Korec and Znám [KZ87] (classes a modulo p^n with $p \nmid a$, for odd primes p such that 2 is a primitive root modulo p^2) and Andaloro [And00] (the class 1 modulo 16). His proof (Lemmas 3.2, 3.6 and 3.8 there) can be read as producing, for every t with $3 \nmid t$, a predecessor of t in every residue class modulo every m, although this is not stated as a theorem. Theorem A extends this to every odd multiplier that is not a Wieferich number, where the role of 2 being a primitive root modulo 3^k is taken by Lemma 6.5, and gives an explicit bound on the least such predecessor. Corollary 3.2 extends Monks's sufficiency theorem to the same multipliers. The question matters because a residue class containing no number that reaches 1 proves at once that Col_q has orbits avoiding 1. By Theorem B and Proposition 5.2 such a class exists for every Wieferich number (defined in §2). Theorem A shows that no such class exists when q is odd and not a Wieferich number, in particular for $q = 3$ and $q = 5$.

Density results and coverage results are logically independent: a set of positive density can miss entire residue classes, and a set meeting every class can have density zero. The nearest density

statement for $3x+1$ is in an unrefereed manuscript of Mazur [Maz26]: the predecessor set of every t with $3 \nmid t$ has positive lower natural density, while for $3 \mid t$ it consists of the numbers $2^k t$, as in Proposition 3.1. Whether these sets have positive density within each residue class is not addressed there.

Theorem A (coverage). *Let $q > 1$, and suppose $2^d = 1 + qu$ for some $d, u \in \mathbb{N}$ with $\gcd(q, u) = 1$. Let $q \le 2^L$, and let $t \ge 1$ satisfy $t \equiv 2^i \pmod q$ for some $i \ge 0$. Then for all $m \ge 1$ and all $r, B \in \mathbb{N}$ there is a w with*

$$w \to t, \qquad w \equiv r \pmod m, \qquad B < w \le (B+1)\, t\, 2^{2m^{L+1}q^m}.$$

Such a w is a *witness* for the residue r modulo m above B; the number t is the *target*, the root of its backward tree.

By Lemma 2.1, the hypothesis on q says exactly that q is odd and not a Wieferich number. The condition on q is a hypothesis, and Theorem B shows it cannot be dropped.

The guiding example. The map Col_5 has three known cycles, with least odd elements 1, 13 and 17. A trajectory that enters one cycle stays there, so the backward trees of these cycles are pairwise disjoint. Since 2 generates $(\mathbb{Z}/5)^\times$, every t with $5 \nmid t$ satisfies the hypothesis of Theorem A:

Corollary 1.1. *If $5 \nmid t$, then for every $m \ge 1$ and every $r, B \in \mathbb{N}$ some $w \equiv r \pmod m$ with $B < w \le (B+1)\, t\, 2^{2^{5m}}$ reaches t under $5x+1$.*

So the backward tree of 1, of 13 and of 17 each meets every residue class modulo every m, effectively, although no number lies in two of them. So coverage of residue classes does not determine which cycle, if any, an orbit enters, and does not imply that every orbit reaches 1.

Theorem B (trap). *Let $g > 1$ divide q, and suppose 2 has the same order d modulo q and modulo qg. If t lies in the set*

$$\{x : x \equiv 2^i \pmod{qg} \text{ for some } i \ge 0\} \;\cup\; \{x : g \mid x\},$$

then so does every x reaching t under Col_q.

Together with Theorem A, this gives a classification for odd q (Proposition 5.2): the predecessors of 1 meet every residue class modulo every m if and only if q is not a Wieferich number. The smallest Wieferich number is 21; for $21x+1$, no $x \equiv 5 \pmod{63}$ reaches 1. Franco and Pomerance [FP95] proved that for every Wieferich number q some orbit of Col_q avoids 1, and that Wieferich numbers have relative density 1 among odd numbers (as stated in [Mor12, §9.7]). Theorem A therefore concerns a set of multipliers that has density zero but is infinite: it contains every power of 3 (§2).

The same results apply to $3x+1$, with bound $(B+1)\, t\, 2^{2^{4m}}$ for every t with $3 \nmid t$ (Corollary 4.2); the coverage statement itself is implicit in the proof of [Mon06], and the bound is what is added. As $5x+1$ shows, they bear on residue classes, not on whether orbits reach 1.

The double exponential in Theorem A is needed by the proof only for the odd prime factors of m that do not divide q. For powers of q the bound is singly exponential, $(B+1)\, t\, 2^{d(m+1)}$ (Proposition 4.3); for $5x+1$ this is $(B+1)\, t\, 16^{m+1}$. When every odd prime factor of m divides q, for instance for powers of 2, the exponent is polynomial in m (Corollary 6.7).

Theorems A and B are also verified with Lean v4.35.0 in the accompanying file; the formal proofs are independent of the arguments given here, and Appendix A lists the formal statements. Everything else in this paper is proved in the text, except where a statement is attributed to the literature or labeled as heuristic.

§2 treats the condition on q, §3 the consequences of Theorem A, §4 the examples, §5 Wieferich numbers and Theorem B, §6 the proof of Theorem A, and §7 its scope.

2 The multiplier

Lemma 2.1. *Let $q > 1$. There are d, u with $2^d = 1 + qu$ and $\gcd(q, u) = 1$ if and only if q is odd and not a Wieferich number.*

Proof. If q is even, $2^d = 1 + qu$ forces $d = 0$, $u = 0$ and $\gcd(q, 0) = q > 1$. Let q be odd, $d_0 = \mathrm{ord}_q(2)$, and $2^{d_0} = 1 + qv$. If $\gcd(q, v) = 1$, take $d = d_0$. Conversely let $2^d = 1 + qu$ with $\gcd(q, u) = 1$. Then $d_0 \mid d$; write $d = d_0 j$. By the binomial theorem $(1+qv)^j = 1+jqv+q^2z$ for some z, so $u = jv+qz$. A prime dividing both q and v would divide u, which is impossible. □

This is the definition of Franco and Pomerance [FP95, Definition 2], also used by Moree [Mor12, §9.7]; the Wieferich numbers in this sense form OEIS sequence A182297 [OEIS]. Other definitions under the same name are in use, for example OEIS A077816; throughout, we mean A182297.

So q is a Wieferich number when some prime $p \mid q$ divides $2^d - 1$ to a higher power than it divides q, where $d = \mathrm{ord}_q(2)$. For a prime q, this is the Wieferich condition $q^2 \mid 2^{q-1} - 1$, met by 1093 and 3511. For a composite it is common, through interaction between prime factors: if a prime p with $p \mid q$ but $p^2 \nmid q$ divides $\mathrm{ord}_r(2)$ for another prime $r \mid q$, then lifting the exponent puts an extra factor p into $2^d - 1$. For $q = 21$: $\mathrm{ord}_7(2) = 3$, so $3 \mid d = 6$, and $2^6 - 1 = 63 = 21 \cdot 3$. The smallest Wieferich number is 21, the first term of A182297.

The numbers $3, 5, 7, 9, 15$ are not Wieferich numbers: $2^2 = 1 + 3$, $2^4 = 1 + 5 \cdot 3$, $2^3 = 1 + 7$, $2^6 = 1 + 9 \cdot 7$, $2^4 = 1 + 15$. More generally $\mathrm{ord}_{3^k}(2) = 2 \cdot 3^{k-1}$, and lifting the exponent gives $v_3\big(4^{3^{k-1}} - 1\big) = 1 + (k - 1) = k$, so no power of 3 is a Wieferich number.

For even q the conclusion of Theorem A fails as well: an odd number maps to an odd number, so an orbit that reaches an odd number other than 1 never reaches 1, and the predecessors of 1 are exactly the powers of 2.

3 The main theorem and its readings

Proposition 3.1 (classification of targets). *Let q be odd, not a Wieferich number, and $t \geq 1$. Either $t \equiv 2^i \pmod q$ for some i, and the predecessors of t meet every residue class modulo every m; or the predecessors of t are exactly the numbers $2^k t$, $k \geq 0$.*

Proof. The first case is Theorem A. Let y reach t, and suppose some term of the orbit before it first reaches t is odd; let x be the last such term. All later terms up to t are obtained by halving, so $qx + 1 = 2^k t$ for some $k \geq 0$. Then $2^k t \equiv 1 \pmod q$, so $t \equiv 2^{-k}$ is a power of 2 modulo q. Hence, if t is not a power of 2 modulo q, every term before t is even and $y = 2^k t$. □

In the second case the predecessors miss the class 1 mod q, so the hypothesis on t in Theorem A cannot be weakened. Since q is odd, $t \equiv 2^i$ implies $\gcd(q, t) = 1$. For $q = 3$ and $q = 5$, where 2 generates all units, the condition is simply $q \nmid t$.

Write $X = t\, 2^{2m^{L+1}q^m}$. Theorem A has four consequences, each proved here in a line or two; none is part of the Lean file.

Infinitude and profinite density. Every interval $(B, (B + 1)X]$ contains a predecessor of t in each residue class modulo m. Applying this with B replaced by a predecessor already found gives a larger one, so every residue class modulo every m contains infinitely many predecessors

of t. The profinite integers $\widehat{\mathbb{Z}}$ are the inverse limit of the rings $\mathbb{Z}/m\mathbb{Z}$, and a set $S \subseteq \mathbb{N}$ is dense in $\widehat{\mathbb{Z}}$ exactly when it meets every residue class modulo every m. So the predecessor set of every t in the first case of Proposition 3.1 is dense in $\widehat{\mathbb{Z}}$; this density is topological, not a proportion.

No congruence obstruction. For t as in Theorem A, a set defined by a finite Boolean combination of congruences is a union of residue classes modulo the least common multiple of the moduli, so if it is nonempty it contains a predecessor of t. No statement of the form "no $w \equiv r \pmod m$ reaches t" is true for such t. In particular, for q odd and not a Wieferich number, no residue class can certify that some orbit of Col_q avoids 1. For every Wieferich number such a class exists (Proposition 5.2), so for the target 1 the condition on q is exactly the boundary of this method.

Sufficiency. Following Monks [Mon06], call a set S of positive integers *sufficient* for Col_q if the orbit of every positive integer meets the orbit of some element of S. Using the accelerated map instead gives the same notion: its orbits omit only the even values $qx+1$, and two orbits sharing such a value also share its half.

Corollary 3.2. *If q is odd and not a Wieferich number, every arithmetic progression $a + b\mathbb{N}$ with $b \geq 1$ is sufficient for Col_q.*

Proof. Let $x \geq 1$. Halving x until it is odd gives an odd y in its orbit, and then $t = qy + 1 \equiv 1 = 2^0 \pmod q$ is in its orbit too. Theorem A with $m = b$, threshold a and $r = a$ gives $w > a$ with $w \equiv a \pmod b$ and $w \to t$, so $w \in a + b\mathbb{N}$ and the orbits of x and w meet at t. □

For $q = 3$ this is Monks's theorem [Mon06, Theorem 1.1], here with an explicit bound on the element of S. For Wieferich q the question is left open: Theorem B does not settle it, since orbits may merge outside the trapped set.

Cycles. Let q be odd. Every element t of every cycle of Col_q is a power of 2 modulo q. Indeed a cycle contains an odd element, since halving alone decreases. Going around the cycle backward from t, let x be the first odd element reached (this may be t itself, after a full turn). The path from x to t is the step $x \mapsto qx + 1$ followed by halvings, so $qx + 1 = 2^k t$ for some $k \geq 0$ and $t \equiv 2^{-k} \pmod q$. For $q = 3$ this says that no element of a cycle is divisible by 3. Hence Theorem A applies to every cycle element:

Corollary 3.3. *If q is odd and not a Wieferich number, then the predecessor set of every element of every cycle of Col_q meets every residue class modulo every m, with the bound of Theorem A.*

Equivalently, every arithmetic progression contains a number whose orbit enters the cycle; for $q = 3$ this is what the proof of [Mon06, Corollary 1.3] shows. The target 1 alone would say nothing about a nontrivial cycle, which is disjoint from the backward tree of 1; the content comes from being free to take t on the cycle.

4 Examples

Proposition 4.1. *For all $m \geq 1$, $2m^4\, 5^m \leq 2^{5m}$ and $2m^3\, 3^m \leq 2^{4m}$.*

Proof. For $m = 1, 2$ the left sides are $10, 800$ and $6, 144$, against $32, 1024$ and $16, 256$. For $m \geq 2$, passing from m to $m+1$ multiplies the left sides by at most $5 \cdot \frac{81}{16} < 32$ and $3 \cdot \frac{27}{8} < 16$. □

Proof of Corollary 1.1. $2^4 = 1 + 5 \cdot 3$ with $\gcd(5, 3) = 1$, and every t with $5 \nmid t$ is a power of 2 modulo 5. Take $L = 3$ and apply Theorem A and Proposition 4.1. □

The cycles of $5x + 1$. Under Col_5,

$$1 \to 6 \to 3 \to 16 \to 8 \to 4 \to 2 \to 1,$$

$$13 \to 66 \to 33 \to 166 \to 83 \to 416 \to 208 \to 104 \to 52 \to 26 \to 13,$$

$$17 \to 86 \to 43 \to 216 \to 108 \to 54 \to 27 \to 136 \to 68 \to 34 \to 17.$$

The cycle through 13 is Crandall's [Cra78; Mor12, §9.7]. The cycle through 17 is recorded by Steiner [Ste81], who shows that the cycles through 1 and 13 are the only ones consisting of a single circuit (one run of odd steps followed by one run of halvings); the cycle through 17 consists of two. By Corollary 1.1 the three backward trees, which are pairwise disjoint (§1), each meet every residue class modulo every m.

Whether Col_5 has further cycles, and whether every orbit of Col_5 enters a cycle, are open.

Corollary 4.2. *If $3 \nmid t$, then for every $m \geq 1$ and every $r, B \in \mathbb{N}$ some $w \equiv r \pmod m$ with $B < w \leq (B+1)\, t\, 2^{2^{4m}}$ reaches t under $3x+1$.*

Proof. $2^2 = 1 + 3$, every t with $3 \nmid t$ is $\equiv 1$ or $2 \pmod 3$, and $L = 2$. Apply Theorem A and Proposition 4.1. □

For moduli that are powers of q, the bound of Theorem A can be made singly exponential.

Proposition 4.3 (powers of q). *Let q be odd and not a Wieferich number, let $2^d = 1 + qu$ with $\gcd(q, u) = 1$, and let $t \geq 1$ satisfy $t \equiv 2^i \pmod q$ for some $i \geq 0$. For every $a \geq 0$, with $m = q^a$, and all $r, B \in \mathbb{N}$, some $w \equiv r \pmod m$ with $B < w \leq (B+1)\, t\, 2^{d(m+1)}$ reaches t.*

Proof. Here $d \geq 1$, since $d = 0$ forces $u = 0$ and $\gcd(q, 0) = q > 1$. As $2^d \equiv 1 \pmod q$ and $t \equiv 2^i$, there is e with $0 \leq e < d$ and $2^e t \equiv 1 \pmod q$. For $k \geq 1$ put $R_k = (2^{e+dk} t - 1)/q$. This is an integer; it is odd, because $e + dk \geq 1$ makes the numerator odd and q is odd; and it is positive. As $qR_k + 1 = 2^{e+dk} t$, one odd step and $e + dk$ halvings take R_k to t.

For $h \geq 1$, $R_{k+h} - R_k = 2^{e+dk} t\, (2^{dh} - 1)/q$, and $2^{e+dk} t$ is prime to q, since $t \equiv 2^i$ and q is odd. So $R_{k+h} \equiv R_k \pmod{q^a}$ if and only if $(1 + qu)^h \equiv 1 \pmod{q^{a+1}}$, which by Lemma 6.4 (with $K = a$) holds if and only if $q^a \mid h$. Hence $R_1, \ldots, R_m$ lie in distinct classes modulo m, so they meet every class. In particular there is $k_r \in \{1, \ldots, m\}$ with $R_{k_r} \equiv r$, and $R_k \equiv r$ exactly when $k \equiv k_r \pmod m$.

The R_k increase without bound. Let k be the least index with $k \equiv k_r \pmod m$ and $R_k > B$, and put $w = R_k$. If $k = k_r$, then $w < 2^{e+dm} t/q < t\, 2^{d(m+1)}$. Otherwise $k - m \geq 1$ and $R_{k-m} \leq B$, and $qw + 1 = 2^{dm}(qR_{k-m} + 1) \leq 2^{dm}(qB + 1)$ gives $w < 2^{dm}(B+1)$. In both cases $w \leq (B+1)\, t\, 2^{d(m+1)}$. □

The proof uses only Lemma 6.4. Proposition 4.3 is proved here and is not part of the Lean file.

Corollary 4.4. *If $5 \nmid t$, then for every $a \geq 0$, with $m = 5^a$, and all $r, B \in \mathbb{N}$, some $w \equiv r \pmod m$ with $B < w \leq (B+1)\, t\, 16^{m+1}$ reaches t under $5x+1$. If $3 \nmid t$, the same holds for $m = 3^a$ under $3x+1$, with 4^{m+1} in place of 16^{m+1}.*

Proof. Take $2^4 = 1 + 5 \cdot 3$ and $2^2 = 1 + 3$ in Proposition 4.3; for $q = 3$ and $q = 5$ every t prime to q is a power of 2 modulo q. □

5 Wieferich numbers

Theorem B is stated in §1; its hypothesis on d (a positive integer) is equivalent to: $2^d \equiv 1 \pmod{qg}$, and every k with $2^k \equiv 1 \pmod q$ is a multiple of d. Call x *trapped* if it lies in the

set of Theorem B. By Theorem B the complement of the trapped set is mapped into itself by Col_q; it is a union of residue classes modulo qg (compare the invariant sets of residue classes in Buttsworth and Matthews [BM90]).

Proof of Theorem B. It suffices to show that x is trapped whenever $\mathrm{Col}_q(x)$ is. If x is even and $x/2 \equiv 2^i$, then $x \equiv 2^{i+1}$; if $g \mid x/2$ then $g \mid x$. If x is odd, then $\mathrm{Col}_q(x) = qx + 1 \equiv 1 \pmod{g}$, so $g \nmid qx + 1$, and therefore $qx + 1 \equiv 2^i \pmod{qg}$ for some i. Reducing modulo q gives $2^i \equiv 1 \pmod{q}$, so $d \mid i$ and $2^i \equiv 1 \pmod{qg}$. Hence $qg \mid qx$, and $g \mid x$. □

Example 5.1. *For $q = 21$, $g = 3$: the powers of 2 modulo 21 are $1, 2, 4, 8, 16, 11$, so $d = 6$, and $2^6 = 64 \equiv 1 \pmod{63}$. The trapped classes modulo 63 are $1, 2, 4, 8, 16, 32$ and the multiples of 3. The class of 5 is neither, so no $x \equiv 5 \pmod{63}$ reaches 1 under $21x + 1$.*

Proposition 5.2 (classification of multipliers). *Let $q > 1$ be odd. The predecessors of 1 under Col_q meet every residue class modulo every m if and only if q is not a Wieferich number.*

Proof. If q is not a Wieferich number, apply Theorem A with $t = 1$. Otherwise let $d = \mathrm{ord}_q(2)$, $2^d = 1 + qv$ and $g = \gcd(q, v) > 1$. Then $qg \mid 2^d - 1$, so 2 has order d modulo qg, and Theorem B applies to $t = 1$, which is trapped. At most $d + q$ classes modulo qg are trapped, and $d + q < 2q < qg$ because $d < q$ and $g \geq 3$. So some class modulo qg contains no predecessor of 1. □

Every element of that class has an orbit avoiding 1. Call q a *Crandall number* if Col_q has an orbit avoiding 1 [Mor12, §9.7]. The argument recovers the theorem of Franco and Pomerance [FP95] that every Wieferich number is a Crandall number. Their proof, like Crandall's argument for 1093 [Cra78], shows that the odd numbers x for which $qx + 1$ is a power of 2 have no odd predecessors. These x are $(2^{kd} - 1)/q$, hence multiples of g, so they lie in the trapped set. Theorem B extends this dead end to an explicit union of residue classes closed under taking predecessors; we do not claim it is new.

The converse fails, and Theorem A explains how. Crandall showed that 5, 181 and 1093 are Crandall numbers [Cra78; Mor12, §9.7]: for 5 and 181 through cycles, with least odd elements 13 and 27. Neither 5 nor 181 is a Wieferich number. So by Theorem A, for these two multipliers no residue class consists entirely of numbers whose orbits avoid 1: such orbits can be exhibited by cycles, but not by congruences.

For a Wieferich number q and a target outside the trapped set, this paper does not determine which residue classes its predecessors meet.

6 Proof of Theorem A

Plan. By Lemma 6.1 below, it suffices to find a word ω of length ℓ with s ones, an exponent $E \geq 0$ and a number $w > B$ with $q^s w + \kappa(\omega) = 2^{\ell+E} t$ and $w \equiv r \pmod{m}$. Write $m = 2^\alpha m_q m'$ as in §6.2. The congruence splits into three parts, each handled by one idea. Modulo 2^α, the first α letters of ω determine w (Lemma 6.2). Modulo m', the two branches of T_q are bijections whose compositions include every translation, so any class can be steered to any other (Lemma 6.3). Modulo powers of q, the powers of 2 fill every class that is 1 modulo q, and this is where the hypothesis that q is not a Wieferich number is used (Lemmas 6.4 and 6.5). The word is chosen first, to meet the conditions modulo 2^α, m' and q^{K+1}; the exponent E is chosen last, to meet the condition modulo $q^{s+\beta}$, which depends on the word. §6.1–6.6 prove the lemmas, §6.7 shows that a witness exists, and §6.8–6.9 bound the least one.

The case $m = 1$ is immediate: take $w = 2^k t$ with $2^{k-1} \leq B < 2^k$, or $k = 0$ when $B = 0$. Assume $m \geq 2$, and fix d, u with $2^d = 1 + qu$, $\gcd(q, u) = 1$; we may take $d = \mathrm{ord}_q(2) \leq q$.

The Lean file certifies the statements: the definitions and hypotheses of Appendix A, and the conclusions of Theorems A and B. The proof below is written for the reader and need not follow the Lean proof step by step; the table gives the nearest Lean declarations for each step.

Step	**Statement**	**Nearest Lean declarations**
Encoding	Lemma 6.1	`integrality`, `offset_append`
Splitting m	§6.2	`split_two_power`, `split_coprime`
2-adic part	Lemma 6.2	`parity_prefix`, `prefix_mod`, `follow_prefix`
Part modulo m'	Lemma 6.3	`residueReach_all`, `boundedReach_all_at`
q-adic part	Lemmas 6.4, 6.5	`lift_exponent`, `order_iff`, `unit_power_cover`
Joining and the exponent	Lemma 6.6, §6.7	`padic_word`, `word_exists`, `exponent_lemma`
Witness and size	§6.7–6.9	`witness_exists`, `witness_size`, `exponent_bound`
Assembly	Theorem A	`predecessor_bound_of_two_le`, `predecessor_bound`

6.1 Parity words

A *word* is a finite sequence of letters 0 and 1. Letter ε acts by $S_\varepsilon(x) = (q^\varepsilon x + \varepsilon)/2$: halving for 0, and $x \mapsto qx + 1$ followed by halving for 1. A word $\omega = \varepsilon_1 \cdots \varepsilon_\ell$ acts by applying S_{ε_1} first. Let $s(\omega)$ be its number of ones and define its *offset* by $\kappa(\varnothing) = 0$ and $\kappa(\varepsilon\,\omega') = q^{\,s(\omega')}\varepsilon + 2\,\kappa(\omega')$. Composing the affine maps gives

$$S_\omega(x) = \frac{q^{s(\omega)}x + \kappa(\omega)}{2^\ell}, \qquad \kappa(\omega\omega') = q^{s(\omega')}\kappa(\omega) + 2^{|\omega|}\kappa(\omega').$$

Lemma 6.1. *If $2^\ell \mid q^s x + \kappa(\omega)$ for a word ω of length ℓ with s ones, then every intermediate value $S_{\varepsilon_1\cdots\varepsilon_i}(x)$ is an integer of the parity ε_{i+1} requires, and $x \to S_\omega(x)$.*

Proof. By induction on ℓ. Divisibility by 2 means $q^{\varepsilon_1}x + \varepsilon_1$ is even, so x has parity ε_1 and $S_{\varepsilon_1}(x) = \mathrm{Col}_q^{1+\varepsilon_1}(x)$; the formula for κ passes the remaining divisibility to $S_{\varepsilon_1}(x)$. □

A parity word is a parity vector in the sense of Terras [Ter76], Everett [Eve77] and Lagarias [Lag85]; Wirsching and Monks use equivalent back-tracing vectors, which record the number of halvings between odd steps, and the affine formula and concatenation rule above are [Wir98, Lemma 2.13, Corollary 2.10] in that language (numbering as cited in [Mon06]). It suffices to find a word ω, an exponent E and a w with

$$q^s w + \kappa(\omega) = 2^{\ell+E}t, \qquad w \equiv r \pmod m,$$

since then $w \to 2^E t \to t$.

6.2 Splitting the modulus

Write $m = 2^\alpha m_q m'$ with m_q, m' odd, every prime of m_q dividing q, and $\gcd(m', q) = 1$. Choose β with $m_q \mid q^\beta$ and $2^\beta \le m_q$; the largest exponent of a prime in m_q works. Let $\mathrm{ord}_{m'}(2) = e_q e'$

with $\gcd(e', q) = 1$ and every prime of e_q dividing q, and choose K likewise with $e_q \mid q^K$ and $2^K \le e_q$. Then $2^{q^K e'} \equiv 1 \pmod{m'}$.

6.3 The 2-adic part

Lemma 6.2. *Let σ be the first α letters of the parity sequence of r under $x \mapsto S_{x \bmod 2}(x)$. Then $2^\alpha \mid q^{s(\sigma)} r + \kappa(\sigma)$. Moreover, if $\omega = \sigma U$ is any word beginning with σ, of length ℓ with s ones, and $2^\ell \mid q^s w + \kappa(\omega)$, then $w \equiv r \pmod{2^\alpha}$.*

Proof. The first claim is $S_\sigma(r) \in \mathbb{N}$. For the second, the concatenation rule gives $q^s w + \kappa(\sigma U) = q^{s(U)}\big(q^{s(\sigma)} w + \kappa(\sigma)\big) + 2^\alpha \kappa(U)$. Since $2^\alpha \mid 2^\ell$ and q is odd, $2^\alpha \mid q^{s(\sigma)} w + \kappa(\sigma)$. Subtracting the first claim gives $q^{s(\sigma)}(w - r) \equiv 0 \pmod{2^\alpha}$, and q is odd. □

6.4 The part modulo m'

Since 2 and q are invertible modulo m', the maps S_0 and S_1, defined on $\mathbb{Z}/m'$ by the same formulas using the inverse of 2, are bijections. The formula for S_ω in §6.1 is an identity over $\mathbb{Z}[1/2]$, so it also holds on $\mathbb{Z}/m'$: a congruence $q^s z + \kappa(\omega) \equiv 2^\ell y \pmod{m'}$ says exactly that $S_\omega(z) = y$ in $\mathbb{Z}/m'$.

Lemma 6.3. *For all $x, y \in \mathbb{Z}/m'$ some word of length at most $m' - 1$ sends x to y. (For $m' = 1$ this is the empty word.)*

Proof. Compositions of bijections of a finite set form a group, so reachability is symmetric. Now

$$S_0 S_1(x) = \frac{qx + 1}{4}, \qquad S_1 S_0(x) = \frac{qx + 2}{4},$$

so $S_1 S_0 \circ (S_0 S_1)^{-1}$ is translation by $1/4$, a unit; its powers are all translations, so every y is reachable from every x. Let R_k be the set reachable from x within k letters. If $R_{k+1} = R_k$, then R_k is closed under S_0 and S_1, hence everything. So $|R_k|$ grows by at least one until $R_k = \mathbb{Z}/m'$, which happens by $k = m' - 1$. □

For $q = 3$, where S_0 and S_1 are $x \mapsto x/2$ and $x \mapsto (3x + 1)/2$, the group they generate modulo m' is determined in [Mon06, Lemma 3.1]: it consists of all maps $x \mapsto 2^a 3^b x + c$, and in particular contains every translation, which is the idea of the proof above.

6.5 The q-adic part

Lemma 6.4. *If q is odd and $\gcd(q, u) = 1$, then $(1 + qu)^{q^K} = 1 + q^{K+1} u_K$ with $\gcd(q, u_K) = 1$. Consequently $(1 + qu)^a \equiv 1 \pmod{q^{K+1}}$ if and only if $q^K \mid a$.*

Proof. Let $j \ge 1$ and $\gcd(q, v) = 1$. By the binomial theorem, $(1 + q^j v)^q = 1 + q^{j+1} v + \binom{q}{2} q^{2j} v^2 + \cdots$. Since q is odd, $q \mid \binom{q}{2}$, so the second term is divisible by q^{2j+1}, and the terms from the third on by q^{3j}; as $j \ge 1$, all are divisible by q^{j+2}. Hence $(1 + q^j v)^q = 1 + q^{j+1} v'$ with $v' \equiv v \pmod{q}$, so $\gcd(q, v') = 1$. Induction gives the first claim.

For the second claim we show, by induction on K, that for all $j \ge 1$ and $\gcd(q, v) = 1$: $(1 + q^j v)^a \equiv 1 \pmod{q^{j+K}}$ if and only if $q^K \mid a$. The case $K = 0$ is trivial, and "if" follows from the first claim. Let $K \ge 1$. Modulo q^{j+1} we have $(1 + q^j v)^a \equiv 1 + a q^j v$, since the other terms are divisible by q^{2j}. So the congruence forces $q \mid av$, hence $q \mid a$. Write $a = qa'$. Then $(1 + q^j v)^a = (1 + q^{j+1} v')^{a'}$, and the claim for $K - 1$ and $j + 1$ gives $q^{K-1} \mid a'$, that is, $q^K \mid a$. The case $j = 1$, $v = u$ is the second claim. (The argument does not use that q is prime. This is

why it proceeds by induction rather than through the order of $1+qu$: for composite q, knowing that this order divides q^K but not q^{K-1} would not by itself determine it.) $\square$

Lemma 6.5. *The powers 2da, $0 \le a < q^K$, are exactly the q^K classes modulo q^{K+1} that are $\equiv 1 \pmod q$.*

Proof. They are distinct by Lemma 6.4 and all $\equiv 1 \pmod q$, and there are exactly q^K such classes. $\square$

Combined with $t \equiv 2^i$: every class modulo q^{K+1} that is a power of 2 modulo q is 2^ν for some $0 \le \nu < dq^K$. For $q = 3$ this is the classical fact that 2 is a primitive root modulo every power of 3, used by Monks [Mon06]; for general q the condition that q is not a Wieferich number is exactly what this step needs.

6.6 Joining the conditions

Lemma 6.6. *For every $z \in \mathbb{N}$ there is a word U, of the form $\omega_0\, 1^{K+1}\, 0^\nu$ with $|\omega_0| \le m' - 1$ and $\nu < dq^K$, such that*

$$q^{s(U)} z + \kappa(U) \equiv 2^{|U|} t \pmod{q^{K+1} m'}.$$

Proof. Modulo q^{K+1}: since $s(U) \ge K+1$, the term $q^{s(U)}z$ vanishes. By the concatenation rule, $\kappa(U) \equiv 2^{|\omega_0|}\kappa(1^{K+1}) \pmod{q^{K+1}}$, because the ones after ω_0 kill $q^{K+1}\kappa(\omega_0)$ and $\kappa(0^\nu) = 0$. So the condition is $\kappa(1^{K+1}) \equiv 2^{K+1+\nu} t$, independent of ω_0. Now $\kappa(1^{K+1}) = \sum_{i=0}^{K} 2^i q^{K-i} \equiv 2^K \pmod q$, so $\kappa(1^{K+1})(2^{K+1}t)^{-1}$ is a power of 2 modulo q, and the remark after Lemma 6.5 gives ν.

Modulo m': the condition says $S_U(z) \equiv t$. With ν fixed, $S_{1^{K+1}0^\nu}$ is a bijection of $\mathbb{Z}/m'$; let y be the preimage of t. Lemma 6.3 gives ω_0 with $S_{\omega_0}(z) \equiv y$. $\square$

6.7 Existence of a witness

Let $q^{s(\sigma)} r + \kappa(\sigma) = 2^\alpha z$ with σ from Lemma 6.2, take U from Lemma 6.6, and put $\omega = \sigma U$, with s ones and length ℓ. Then $Y = q^s r + \kappa(\omega) = 2^\alpha\big(q^{s(U)} z + \kappa(U)\big)$ satisfies $Y \equiv 2^\ell t \pmod{q^{K+1} m'}$. We seek E with

$$2^{\ell+E} t \equiv Y \pmod{q^{s+\beta} m'}. \tag{$*$}$$

Modulo m' this holds whenever $\operatorname{ord}_{m'}(2) \mid E$. Modulo $q^{s+\beta}$, let $j = s + \beta - 1$, so $j \ge K$, and let $y \equiv Y(2^\ell t)^{-1} \pmod{q^{j+1}}$; then $y \equiv 1 \pmod{q^{K+1}}$. By Lemma 6.5 at level j, the order of 2 modulo q^{j+1} is dq^j, and $2^E \equiv y$ exactly for E in one class $E_0 \bmod dq^j$; by Lemma 6.4, $dq^K \mid E_0$. The two conditions are compatible: $\operatorname{ord}_{m'}(2) = e_q e'$ with $e_q \mid q^K$ and $\gcd(e', q) = 1$, so $\gcd(dq^j, e_q e')$ divides dq^K, which divides E_0. By the Chinese remainder theorem, $(*)$ holds for every E in one class modulo

$$\Lambda = \operatorname{lcm}\big(dq^j, \operatorname{ord}_{m'}(2)\big) = \operatorname{lcm}(q^K e',\, dq^j),$$

the two expressions agreeing because $e_q \mid q^K \mid q^j$. In particular $(*)$ holds for arbitrarily large E.

For such E, $(*)$ gives $q^s \mid 2^{\ell+E} t - \kappa(\omega)$, and since $\kappa(\omega) \le q^s 2^\ell$, the number

$$w = \frac{2^{\ell+E}t - \kappa(\omega)}{q^s}$$

is an integer, and it exceeds B as soon as $2^{\ell+E} \geq q^s(B+1+2^\ell)$, which holds for all large E in the class. From $(*)$, $q^s(w-r) \equiv 0 \pmod{q^{s+\beta}m'}$, so $w \equiv r \pmod{q^\beta m'}$, and $m_q \mid q^\beta$ gives $w \equiv r \pmod{m_q m'}$. Lemma 6.2 gives $w \equiv r \pmod{2^\alpha}$, so $w \equiv r \pmod{m}$, and Lemma 6.1 gives $w \to 2^E t \to t$. This proves Theorem A without the bound.

6.8 Making it effective

Since $2^\ell \leq q^s(B+1+2^\ell)$, there is $h \geq 0$ with $q^s(B+1+2^\ell) \leq 2^{\ell+h} < 2q^s(B+1+2^\ell)$. Let E be the least element with $E \geq h$ of the class modulo Λ found in §6.7, so $E \leq h + \Lambda - 1$, and define w as there. Then

$$w \geq \frac{q^s(B+1+2^\ell) - q^s 2^\ell}{q^s} = B+1, \qquad w \leq \frac{2^{\ell+E}t}{q^s} < 2^\Lambda(B+1+2^\ell)\,t \leq (B+1)\,t\,2^{\Lambda+\ell+1}.$$

6.9 The size of the bound

We use four elementary facts. (i) An order is at most its modulus, so $d \leq q$, $e_q \leq e_q e' = \mathrm{ord}_{m'}(2) \leq m'$ and $e' \leq m'$. (ii) $|\omega_0| \leq m'-1$ and $\nu < dq^K$, so $\ell \leq \alpha + m' + K + dq^K$ and $s \leq \alpha + m' + K$, hence $j+1 = s+\beta \leq \alpha+\beta+m'+K$. (iii) $2^{\alpha+\beta+K} \leq 2^\alpha m_q e_q \leq m$, and since $q \leq 2^L$, every x with $2^x \leq m$ has $q^x \leq m^L$; in particular $q^{\alpha+\beta+K} \leq m^L$ and $q^K \leq m^L$. (iv) Both $q^K e'$ and dq^j divide $e'dq^j$, because $K \leq j$; so $\Lambda \leq e'dq^j$.

Hence

$$\Lambda \leq e'dq^j \leq m'\,q^{j+1} \leq m' \cdot q^{\alpha+\beta+K} \cdot q^{m'} \leq m^{L+1}q^m.$$

For the length, $\alpha + K \leq \log_2 m \leq m$ and $m' \leq m$, so $\ell + 1 \leq 3m + 1 + qm^L$. Since $q \geq 3$ we have $L \geq 2$, and $mq^m \geq 2q^2 \geq 6q$, so $m^{L+1}q^m \geq 6qm^L \geq qm^L + 3m + 1$. Thus $\ell + 1 \leq m^{L+1}q^m$ as well, and $\Lambda + \ell + 1 \leq 2m^{L+1}q^m$, which proves Theorem A.

Remark (explanation; not used in the proof). The double exponential has one source: the word ω_0 of Lemma 6.3 may need up to about m' odd steps, and each odd step raises the q-adic precision that the exponent E must cover by a factor q. Reducing it would require reaching every class modulo m' with far fewer odd steps. When $m' = 1$ no such steps are needed, and the same computation gives a much smaller bound.

Corollary 6.7 (moduli without other odd primes). *Let $d = \mathrm{ord}_q(2)$, and let $m = 2^\alpha m_q$, so that every odd prime factor of m divides q. Then for all $r, B \in \mathbb{N}$ some $w \equiv r \pmod{m}$ with*

$$B < w < (B+1)\,t\,2^{\,d\,m^{\log_2 q} + \alpha + d + 1}$$

reaches t.

Proof. For $m = 1$ see the start of §6. Otherwise $m' = 1$, so $\mathrm{ord}_{m'}(2) = 1$, $K = 0$ and $e' = 1$. In Lemma 6.6, ω_0 is empty and $U = 1\,0^\nu$ with $\nu < d$, so $\ell \leq \alpha + d$ and $s \leq \alpha + 1$. Hence $j = s + \beta - 1 \leq \alpha + \beta$ and $\Lambda = dq^j \leq dq^{\alpha+\beta}$. Now $q^\alpha = (2^\alpha)^{\log_2 q}$, and $2^\beta \leq m_q$ gives $q^\beta \leq m_q^{\log_2 q}$, so $q^{\alpha+\beta} \leq m^{\log_2 q}$. §6.8 gives $w < (B+1)\,t\,2^{\Lambda+\ell+1}$. □

The exponent is polynomial in m: for $3x+1$ and $m = 2^\alpha$ it is at most $2m^{\log_2 3} + \alpha + 3$. For m a power of a prime q it is linear, though Proposition 4.3 is sharper there. For $\alpha > 0$ the superlinear factor comes, in this proof, from the prefix σ of Lemma 6.2: it is fixed by $r \bmod 2^\alpha$, may consist of α ones, and each one costs a factor q of precision. The witnesses of Proposition 4.3 cannot help here: they are one odd step from t, and for large k all of them lie in the single class $-q^{-1} \bmod 2^\alpha$, so reaching other classes modulo 2^α needs a longer path, which is what σ provides. This cost is forced only for witnesses of the form used here, a word followed by halvings; it is not a lower bound. We do not know whether an exponent linear in m holds for these moduli, in particular for powers of 2. Corollary 6.7 is proved here and is not part of the Lean file.

7 Scope

Theorem A is a statement about the backward tree, residue class by residue class. It does not bound stopping times, exclude nontrivial cycles, or exclude divergent orbits, and §4 shows that it holds for $5x+1$, where cycles avoiding 1 exist. Lower bounds on the length of nontrivial $3x+1$ cycles, obtained from linear forms in logarithms [Ste78, SdW05], come from size estimates, and the results of [Tao22] concern how small orbits become. Theorem A neither implies nor contradicts them.

The double exponential in Theorem A comes only from the odd prime factors of m that do not divide q (§6.9). For powers of q, Proposition 4.3 gives a single exponential, $(B+1)\,t\,2^{d(m+1)}$. When m has no other odd prime factors, Corollary 6.7 gives an exponent polynomial in m, of degree $\log_2 q$. Both are proved in the text, not in Lean. We do not know whether a linear exponent holds for mixed moduli or for powers of 2, or whether moduli with other odd prime factors require a double exponential.

The bounds are bounds on this proof, not estimates of the least witness. As a heuristic observation, not used anywhere: for $5x+1$, $t = 1$ and $B = 0$, a computation of the backward tree of 1 up to 2^{34} finds a predecessor in every class modulo 2^α for $\alpha \le 8$ and modulo 5^a for $a \le 4$, all below 2^{22}, whereas Proposition 4.3 gives about 2^{2500} for $m = 625$. This suggests that the least witness may grow only polynomially in m; we make no claim to that effect.

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Appendix A. Formal statements

The file is checked with Lean v4.35.0 and uses only its core library. Names in backticks are the file's own. The two main statements use the paper's symbols; the table records the remaining differences.

Lean	**Paper**	**Meaning**
`q`, `m`, `t`, `w`, `x`, `d`, `g`, `r`, `B`, `L` in `predecessor_bound` and `trapped_of_iter`	q, m, t, w, x, d, g, r, B, L	as in the paper
`p` elsewhere in the file	q	the multiplier (a section variable)
`Q` in `Trapped`	qg	the modulus of the trapped set

`collatz` is Col_q. In the definitions below the multiplier is the section variable `p`, so `collatz` and

`Reaches` take it as their first explicit argument, as in `Reaches q w t`, which says that `w` reaches `t`; `iter` is generic in the map. `NonWieferich q` is the hypothesis of Theorem A, equivalent by Lemma 2.1 to q being odd and not a Wieferich number. The definitions:

```
def collatz (n : Nat) : Nat := if n % 2 = 0 then n / 2 else p * n + 1
def iter (f : Nat → Nat) : Nat → Nat → Nat
  | 0, n => n
  | k + 1, n => iter f k (f n)
def Reaches (n v : Nat) : Prop := ∃ k, iter (collatz p) k n = v
class NonWieferich (p : Nat) : Prop where
  one_lt : 1 < p
  unit : ∃ d u, 2 ^ d = 1 + p * u ∧ Nat.Coprime p u
```

Theorem A (coverage), `predecessor_bound`:

```
theorem predecessor_bound (q : Nat) [NonWieferich q] (L : Nat) (hqL : q ≤ 2 ^ L)
    (m : Nat) (hm : 1 ≤ m) (t : Nat) (ht0 : 1 ≤ t)
    (ht : ∃ i, t % q = 2 ^ i % q) (r B : Nat) :
    ∃ w : Nat, B < w ∧ Reaches q w t ∧ w % m = r % m ∧
      w ≤ (B + 1) * t * 2 ^ (2 * m ^ (L + 1) * q ^ m)
```

Theorem B (trap), `trapped_of_iter`, with the trapped set and the hypothesis on d written out:

```
def Trapped (Q g x : Nat) : Prop := (∃ k, x % Q = 2 ^ k % Q) ∨ x % g = 0

theorem trapped_of_iter (q g d t : Nat) (hq : 0 < q) (hg : 1 < g) (hgq : g ∣ q)
    (hQ : 2 ^ d % (q * g) = 1 % (q * g))
    (hmin : ∀ k, 2 ^ k % q = 1 % q → d ∣ k)
    (ht : Trapped (q * g) g t) : ∀ k x, iter (collatz q) k x = t → Trapped (q * g) g
     x
```